



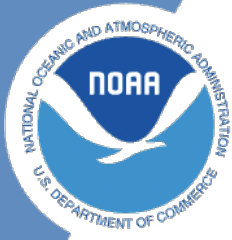
NOAA Technical Memorandum NMFS-XXX-##

NOAA quarto book

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U.S. DEPARTMENT OF COMMERCE
National Oceanic and Atmospheric
Administration
National Marine Fisheries Service
Northwest Fisheries Science Center



**NOAA
FISHERIES**

NOAA quarto book

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Table of contents

1	Introduction	1
1.1	Details	1
2	NOAA Fisheries landings data	2
2.1	Data access and upload	2
2.2	Summarize the landings	3
2.3	Plot the summary	5
2.4	Updated Gulf/Atlantic parsing of data	6
2.5	Plots with updated data	9
3	International landings	11
4	Analysis of international landings estimates from Sea Around Us database	12
4.1	Data access and upload	12
4.2	Read in the high seas landings	12
4.3	Summarize the high seas landings	22
4.4	Read in the landings within exclusive economic zones (EEZs)	24
4.5	Separating the EEZs by area	44
4.6	Summarize catch by regional EEZs	52
4.7	Merge high seas with catch with exclusive economic zones	56
4.8	Plot Western Atlantic catches	60
5	SST	62
6	Analysis of ocean temperature trends	63

List of Figures

List of Tables

1 Introduction

Here we will be displaying various data summaries and analyses related to dolphin management in the U.S. South Atlantic.

1.1 Details

The documentation of this data exploration was developed in R, using R Studio and Quarto.

2 NOAA Fisheries landings data

Here we will explore commercial and recreational landings data accessed through the FOSS (Fisheries One Stop Shop) data portal. The advantage of this portal is that commercial and recreational data are available side-by-side in the same units (pounds landed).

2.1 Data access and upload

Data are downloaded from the NOAA Fisheries FOSS site. For the Data Set, we choose both commercial and recreational. For the Years, we select all years from 1980 to the present. For NMFS Region, we select the "NMFS Regions" button and then select the Gulf, South Atlantic, Middle Atlantic and New England from the box. For the Species, we select "Dolphinfish" and "Dolphinfish**." For Report Format, we choose TOTALS BY YEAR/REGION. Hitting the "RUN REPORT" button then generates the data needed for the subsequent analysis, which can be downloaded as a .csv file under the "Actions" button.

We then read in the data file and clean it for analysis (removing the commas from the numbers so that R recognizes the values as numbers). We output the file header to make sure it was read in and that the numbers were converted correctly.

```
# clear workspace
rm(list = ls())

# read in data file
d <- read.table("data/FOSS_landings.csv", header = T, sep = ",", skip = 1)
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	1980	Gulf	94,984	55,676	Commercial	43
2	1980	Middle Atlantic	400	256	Commercial	0
3	1980	New England	200	54	Commercial	0

2 NOAA Fisheries landings data

4	1980	South Atlantic	76,042	45,852	Commercial	34
5	1981	Gulf	73,075	46,951	Commercial	33
6	1981	Gulf	1,559,544		Recreational	707

```
# take commas out of file -----
d$Pounds <- as.numeric(gsub(",", "", d$Pounds))
d$Dollars <- as.numeric(gsub(",", "", d$Dollars))
d$Metric.Tons <- as.numeric(gsub(",", "", d$Metric.Tons))
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	1980	Gulf	94984	55676	Commercial	43
2	1980	Middle Atlantic	400	256	Commercial	0
3	1980	New England	200	54	Commercial	0
4	1980	South Atlantic	76042	45852	Commercial	34
5	1981	Gulf	73075	46951	Commercial	33
6	1981	Gulf	1559544	NA	Recreational	707

2.2 Summarize the landings

We then summarize the landings (in pounds) by year, sector, and region. The result is a new data frame with each row representing a year and each column representing the total landings for each region and sector.

```
# compile by year for sectors and regions -----
d$Collection <- factor(d$Collection, levels = c("Recreational", "Commercial"))
tab <- tapply(d$Pounds, list(d$Year, d$Collection, d$Region.Name), sum, na.rm = T)
tab[is.na(tab)] <- 0
#head(tab)

dat <- data.frame(cbind(tab[, , "Gulf"], tab[, , "South Atlantic"],
                        tab[, , "Middle Atlantic"], tab[, , "New England"]))
names(dat) <- c("rec_Gulf", "com_Gulf", "rec_SA", "com_SA", "rec_MA", "com_MA", "rec_NE", "com_NE")
dat
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
1980	0	94984	0	76042	0	400	0	200
1981	1559544	73075	8421898	59037	0	0	0	0

2 NOAA Fisheries landings data

1982	4197503	162457	7573035	143960	2097	200	0	0
1983	2162936	190808	9134022	127986	45927	1400	0	400
1984	1189573	282976	7265430	158901	0	1700	0	400
1985	2542089	269477	6684821	142684	113666	5000	0	4800
1986	5540003	492212	6190187	189747	125653	4200	0	200
1987	4209638	403703	6883546	212905	121433	13400	0	1100
1988	1014312	491610	3134070	212709	130240	26600	2196	17800
1989	12551461	1029628	13273504	426392	1386752	81700	437	15300
1990	13389540	1098819	9283973	514980	624928	69106	2714	14233
1991	13760600	1674124	14756442	650333	722090	90722	6294	9816
1992	12481422	714369	5893700	326831	860112	74423	1411	10315
1993	6873820	596967	7077105	520512	3001335	99420	200927	27069
1994	5448709	679903	10554836	618614	1400635	101044	12586	12328
1995	11142952	1177325	14318412	1164608	1096473	198425	24061	13811
1996	7086006	1047020	9128895	548682	1077364	43581	0	7788
1997	19958088	989648	12284366	877303	1422057	105559	0	13265
1998	9324186	422501	7312090	406368	2920384	86502	120322	11553
1999	7329953	612445	13973364	504263	1005852	98927	4458	5990
2000	5641491	582576	16049049	537077	1433991	36048	0	6545
2001	7355385	368503	14945284	470897	477081	51342	5189	29134
2002	3513226	290601	13794853	436472	1295470	83534	600948	14836
2003	4481937	310836	12112718	459351	929763	46851	129513	21275
2004	2402315	435341	9707678	555973	427145	31916	0	29042
2005	2537132	207695	10299670	376989	437472	19703	0	19982
2006	3290474	224668	12805952	433121	887684	27724	0	19645
2007	3754513	370397	13164377	740255	241212	79832	19542	44707
2008	3937153	384954	10417536	627939	264017	57764	0	16328
2009	3860527	496701	13251102	1007355	1538112	55271	0	92010
2010	1216198	111790	8543542	572031	813738	35869	0	14939
2011	3023914	309324	10774812	543811	2173136	15376	187289	26587
2012	3229090	146256	9983276	704240	755558	44918	26892	46171
2013	5425636	187966	8483569	583556	302421	41936	226988	18564
2014	3074373	201943	11925345	1041856	2129222	58868	183673	61505
2015	6286577	154133	14821772	920454	4501793	36190	1198628	37808
2016	1777348	37411	12357817	828133	2212513	61336	16499	15747
2017	2064106	65141	9457460	532973	1820541	16179	48607	30969
2018	1626095	56188	11713866	435059	3104018	20514	97107	12667
2019	3567170	83956	7967978	563132	1518629	48224	66	16918
2020	1952241	26332	5653294	260311	679556	14040	0	1880
2021	2893817	26570	6061024	178822	1051785	25318	123038	11421
2022	1664771	15906	6845573	182798	708494	17360	466897	9131

2 NOAA Fisheries landings data

2023 1427518 13305 6630236 125302 734866 28467 122550 8310

2.3 Plot the summary

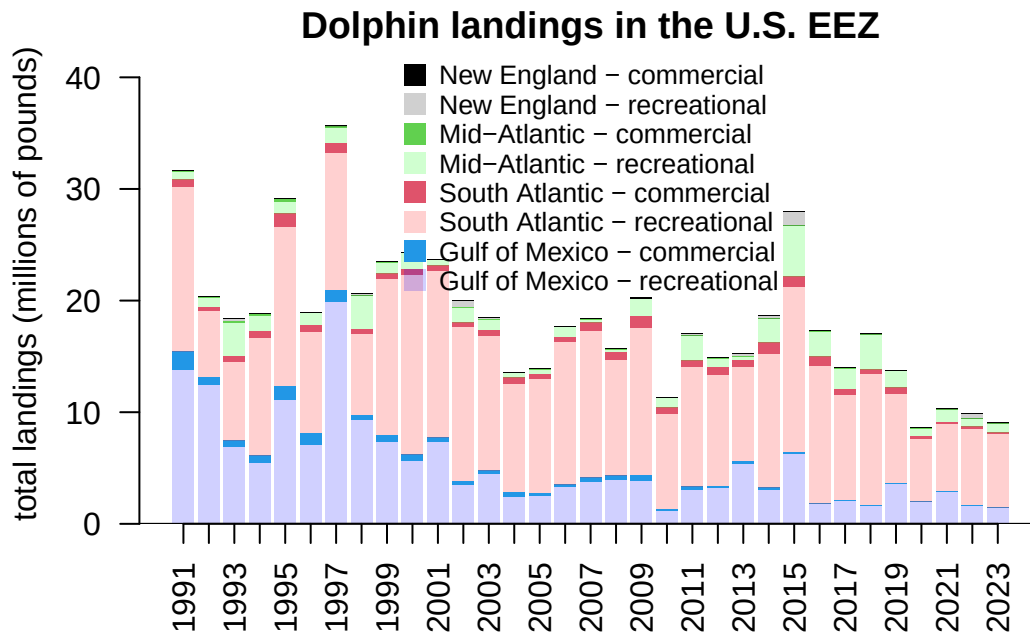
Now we plot the data, using similar color schemes by region. Because the recreational data have high imprecision prior to the 1990s we plot from 1990 to the last year of data.

```
# select years -----
dat <- dat[which(rownames(dat) > 1990), ]

# define colors and plot -----
cols <- c("#0000FF30", 4, "#FF000030", 2, "#00FF0030", 3, "#00000030", 1)

par(mar = c(4, 4, 1, 0), mgp = c(2.5, 1, 0))
yrs <- as.numeric(rownames(dat))

b <- barplot(t(dat)/10^6, names.arg = yrs, col = cols, ylim = c(0, 43),
             border = NA, las = 2, ylab = "total landings (millions of pounds)",
             main = "Dolphin landings in the U.S. EEZ")
legend("top", c("New England - commercial", "New England - recreational",
               "Mid-Atlantic - commercial", "Mid-Atlantic - recreational",
               "South Atlantic - commercial", "South Atlantic - recreational",
               "Gulf of Mexico - commercial", "Gulf of Mexico - recreational"),
       col = cols[8:1], pch = 15, pt.cex = 1.5, bty = "n", y.intersp = 0.9, cex = 0.85)
axis(1, at = b, lab = rep("", length(b)), pos = 0)
abline(h=0)
```



2.4 Updated Gulf/Atlantic parsing of data

There are some issues with the way that the FOSS system splits out Gulf and Atlantic catches by Monroe County, which is on the border between the two jurisdictions. We received data from Mike Larkin (SERO) which has the correct split out of data for Monroe County between the Gulf and Atlantic. We compare the total recreational catch for the Gulf and Atlantic between the FOSS data and the updated data. The numbers are very close; the average difference from the FOSS data is 2%.

```
ml <- read.table("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", header = T, sep = ",")
ml <- ml[which(ml$Year > 1990), ]

# select same years for FOSS data set
dat <- dat[which(yrs <= max(ml$Year)), ]

# check that years match
summary(rownames(dat) == ml$Year)
```

Mode TRUE

2 NOAA Fisheries landings data

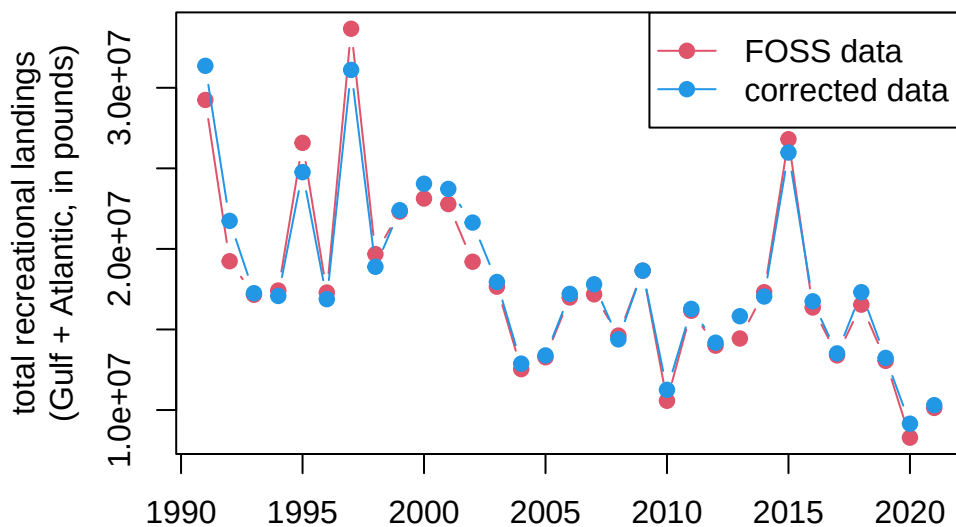
logical 31

```
yrs <- as.numeric(rownames(dat))

# check that totals match
old <- dat$rec_Gulf + dat$rec_SA + dat$rec_MA + dat$rec_NE
new <- ml$GOM + ml$ATL

# plot totals
par(mar = c(4, 6, 2, 1), mgp = c(2.5, 1, 0))
matplot(yrs, cbind(old, new), xlab = "", lty = 1, col = c(2, 4), type = "b", pch = 19,
        ylab = "total recreational landings\n(Gulf + Atlantic, in pounds)",
        main = "Comparison of FOSS data with data corrected for Monroe County")
legend("topright", c("FOSS data", "corrected data"), col = c(2, 4), pch = 19, lty = 1)
```

Comparison of FOSS data with data corrected for Monroe



```
# average % difference between FOSS and updated data
mean((old - new)/old*100)
```

[1] -2.023886

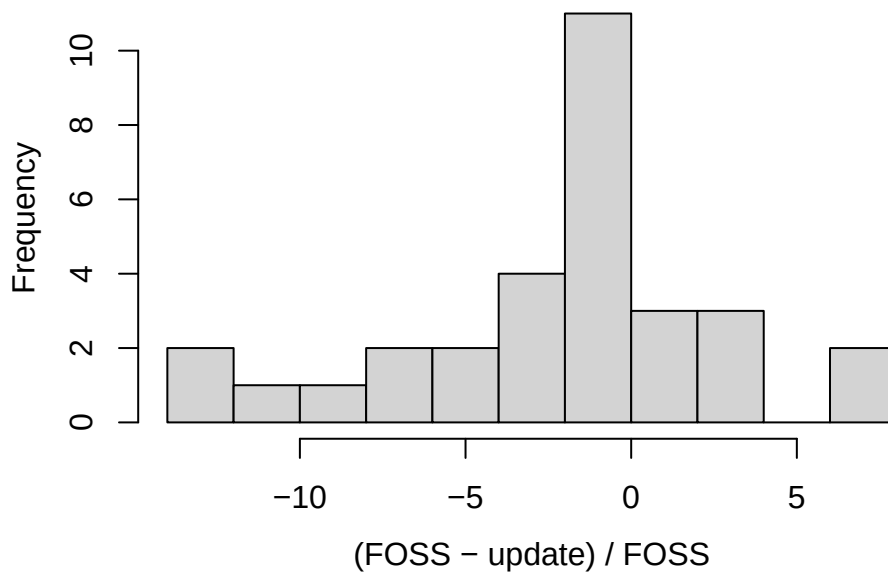
2 NOAA Fisheries landings data

```
# min and max % difference between FOSS and updated data  
range((old - new)/old*100)
```

```
[1] -13.008552  7.587789
```

```
hist((old - new)/old*100, main = "Histogram of percent difference from FOSS data", breaks =  
      xlab = "(FOSS - update) / FOSS")
```

Histogram of percent difference from FOSS data



```
# replace FOSS data with update data  
datnew <- dat  
datnew$rec_Gulf <- ml$GOM  
datnew$rec_SA <- ml$ATL - (datnew$rec_MA + datnew$rec_NE)  
  
# old data set  
tail(dat)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	1777348	37411	12357817	828133	2212513	61336	16499	15747
2017	2064106	65141	9457460	532973	1820541	16179	48607	30969

2 NOAA Fisheries landings data

2018	1626095	56188	11713866	435059	3104018	20514	97107	12667
2019	3567170	83956	7967978	563132	1518629	48224	66	16918
2020	1952241	26332	5653294	260311	679556	14040	0	1880
2021	2893817	26570	6061024	178822	1051785	25318	123038	11421

```
# updated data set  
tail(datnew)
```

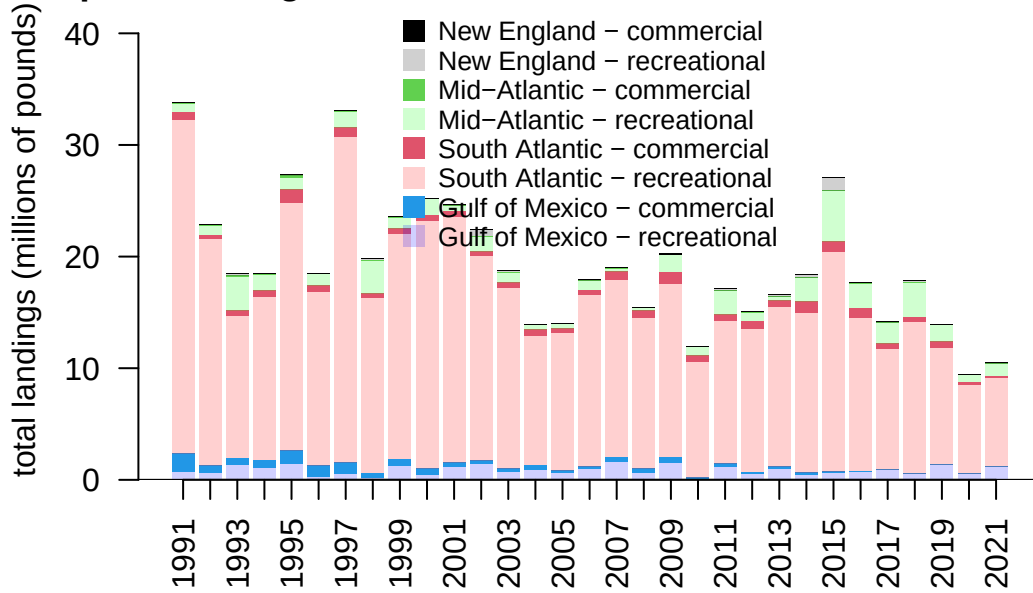
	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	751499	37411	13766851	828133	2212513	61336	16499	15747
2017	868489	65141	10777104	532973	1820541	16179	48607	30969
2018	508570	56188	13603875	435059	3104018	20514	97107	12667
2019	1304865	83956	10410599	563132	1518629	48224	66	16918
2020	555695	26332	7916891	260311	679556	14040	0	1880
2021	1174412	26570	7950716	178822	1051785	25318	123038	11421

2.5 Plots with updated data

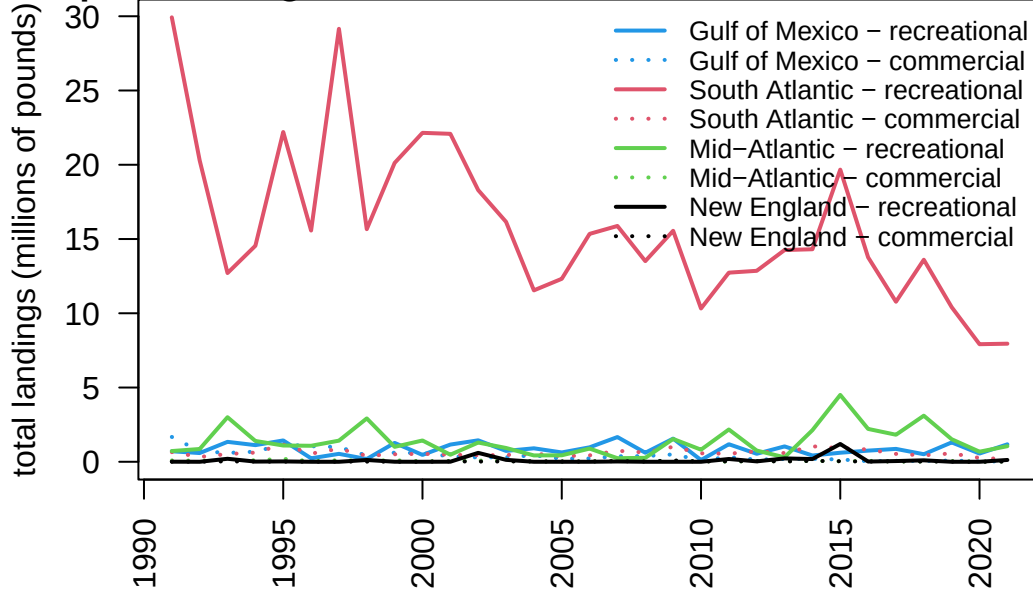
We replace the FOSS values for Gulf and South Atlantic recreational catches with these new values and re-plot the full data set. We can also plot the landings as a line graph so that the relative magnitudes of catch across region and sector are more easily compared.

2 NOAA Fisheries landings data

Dolphin landings in the U.S. EEZ -- correct Monroe Coun



Dolphin landings in the U.S. EEZ -- correct Monroe Coun



3 International landings

4 Analysis of international landings estimates from Sea Around Us database

Here we will explore commercial and recreational landings data as estimated by the Sea Around Us Project. This data set includes reported data for fisheries worldwide as well as reconstructed estimates for unreported landings. For each country, project scientists assemble available data on landings and use additional sociological and fishing data plus expert information and opinion, to produce their best estimates of total landings from all fishing sectors and for all species globally.

4.1 Data access and upload

For catches within each country's exclusive economic zones, data are downloaded from the Sea Around Us site. In the Biodiversity by Taxon Tools and Data, we filter by EEZ for All regions, and then filter by Taxon at the Species level for Common dolphinfish (*Coryphaena hippurus*). After clicking "View graphs" one can access the full data set using the "Download Data" button. The high seas data are accessed in the Basic Search High seas section of the site. The individual regions can be clicked on (e.g., Atlantic, Western Central) and then the data are downloaded using the Download Data button.

The current version of data in use is version 50.1, accessed June 2024. This was still the most recent version of data available as of June 2025.

4.2 Read in the high seas landings

First we read in the high seas landings for the Western Central Atlantic and Northwest Atlantic zones. These are equivalent to FAO areas 21 and 31.

4 Analysis of international landings estimates from Sea Around Us database

```
# clear workspace
rm(list = ls())

install.packages("pals")
library(pals)

# find data files
hslis <- dir("data/SAU/")[grep("HighSeas", dir("data/SAU/"))]

# identify areas 21 and 31
print("File names:")
```

```
[1] "File names:"
```

```
hslis[1]
```

```
[1] "SAU HighSeas 21 v50-1.csv"
```

```
hslis[3]
```

```
[1] "SAU HighSeas 31 v50-1.csv"
```

```
#download data
h1 <- read.csv(paste0("data/SAU/", hslis[1]))
h3 <- read.csv(paste0("data/SAU/", hslis[3]))

# confirm correct areas
print("Areas:")
```

```
[1] "Areas:"
```

```
unique(h1$area_name)
```

```
[1] "Atlantic, Northwest"
```

4 Analysis of international landings estimates from Sea Around Us database

```
unique(h3$area_name)
```

```
[1] "Atlantic, Western Central"
```

```
hs <- rbind(h1, h3)
```

We extract the data for dolphinfish and look at the characteristics of the landings. Most of the landings are in the Western Central Atlantic, and the primary fishing entities are Venezuela and Saint Lucia. The fishing sector is all industrial, and the vast majority of the catch is reported landings, with only a tiny fraction of discards or unreported landings listed in the database. The gear type is primarily long line.

```
# check species name and filter out dolphinfish
unique(hs$common_name)[grep("dolphin", unique(hs$common_name))]
```

```
[1] "Common dolphinfish"
```

```
hs <- hs[which(hs$common_name == "Common dolphinfish"), ]
```

```
# look at the categories within each column of data
apply(hs[1:13], 2, table)
```

\$area_name

	Atlantic,	Northwest Atlantic,	Western Central
	113		160

\$area_type

high_seas
273

\$year

1985	1986	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
1	1	1	1	1	1	1	6	4	1	5	10	5	8	10	12
2013	2014	2015	2016	2017	2018	2019									
24	22	30	33	33	29	34									

4 Analysis of international landings estimates from Sea Around Us database

\$scientific_name

Coryphaena hippurus
273

\$common_name

Common dolphinfish
273

\$functional_group

Large pelagics (≥ 90 cm)
273

\$commercial_group

Perch-likes
273

\$fishing_entity

Barbados	Belize
6	3
Brazil	Canada
19	18
Guyana	Panama
1	1
Saint Lucia	Saint Vincent & the Grenadines
5	28
Spain	Suriname
80	1
Trinidad & Tobago	USA
14	74
Venezuela	
23	

\$fishing_sector

Industrial

4 Analysis of international landings estimates from Sea Around Us database

273

\$catch_type

Discards	Landings
5	268

\$reporting_status

Reported	Unreported
253	20

\$gear_type

artisanal fishing gear	bottom trawl	gillnet
7	9	11
hand lines	longline	mixed gear
17	187	1
pole and line	pots or traps	unknown by source
19	9	13

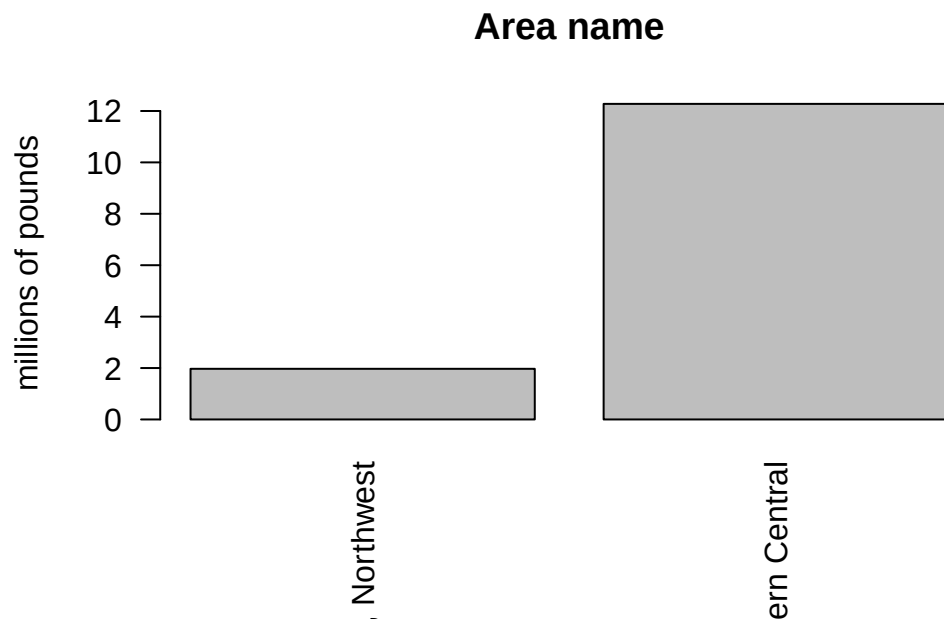
\$end_use_type

	Direct human consumption	Fishmeal and fish oil
5	194	37
Other		
37		

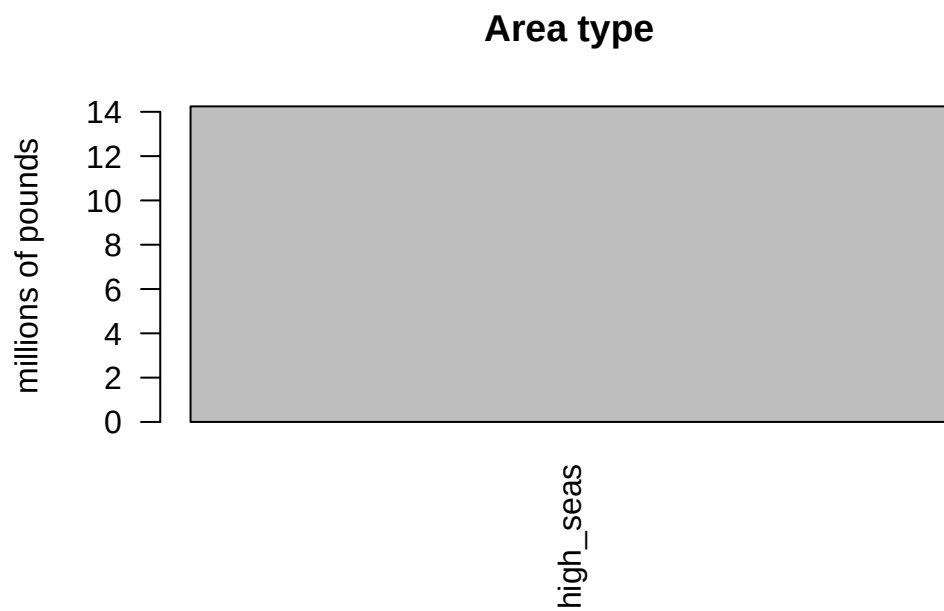
```
# convert from metric tonnes to pounds
hs$pounds <- hs$tonnes * 1000 * 2.20462

# produce barplot of key data fields
#par(mar = c(10, 5, 3, 1), mfrow = c(4, 2))
barplot(tapply(hs$pounds/10^6, hs$area_name, sum, na.rm = T), las = 2, ylab = "millions of
```

4 Analysis of international landings estimates from Sea Around Us database

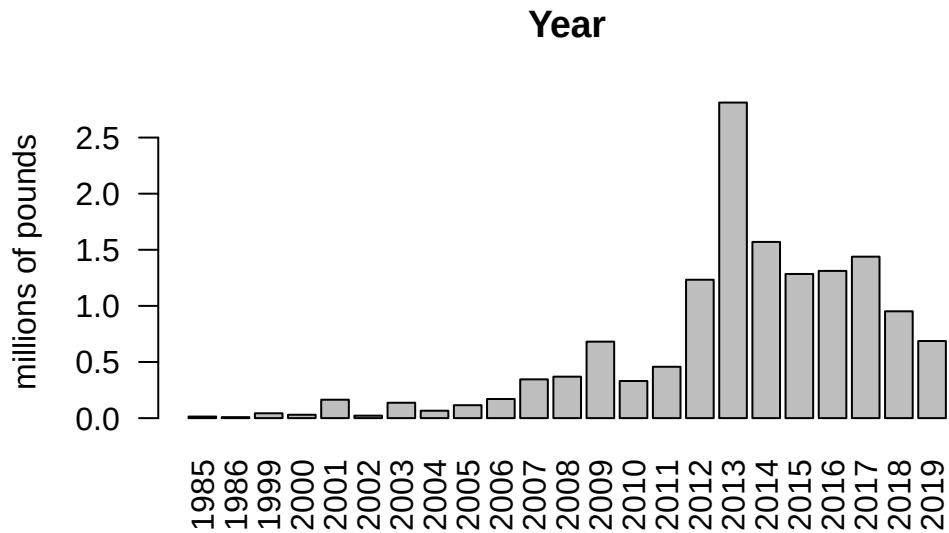


```
barplot(tapply(hs$pounds/10^6, hs$area_type, sum, na.rm = T), las = 2, ylab = "millions of pounds")
```



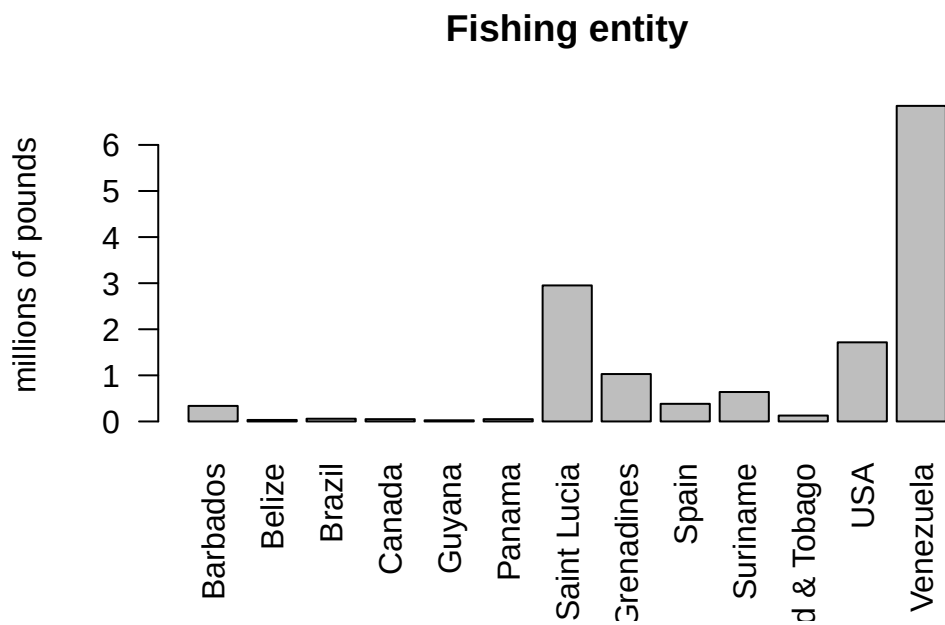
4 Analysis of international landings estimates from Sea Around Us database

```
barplot(tapply(hs$pounds/10^6, hs$year, sum, na.rm = T), las = 2, ylab = "millions of pounds")
```

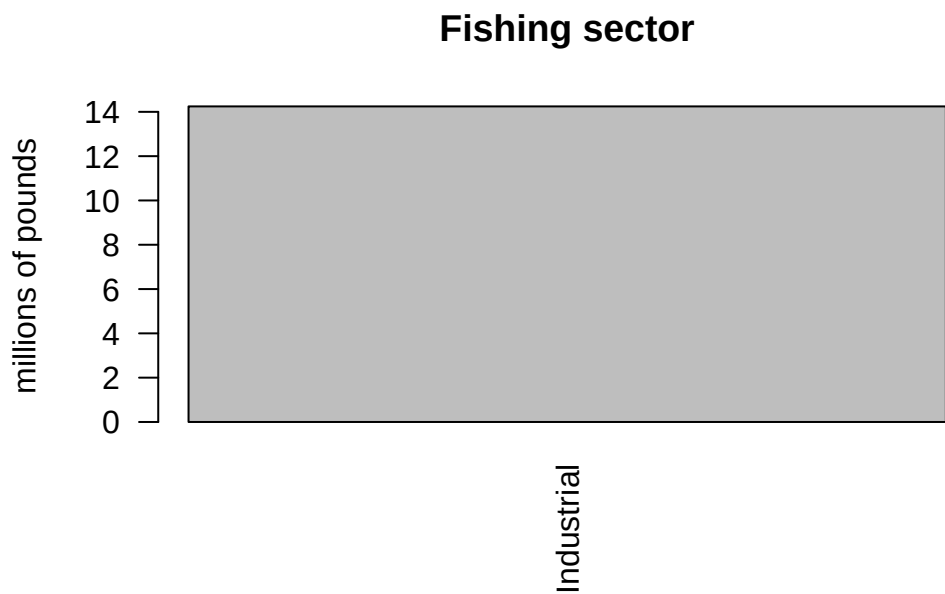


```
barplot(tapply(hs$pounds/10^6, hs$fishing_entity, sum, na.rm = T), las = 2, ylab = "million")
```

4 Analysis of international landings estimates from Sea Around Us database

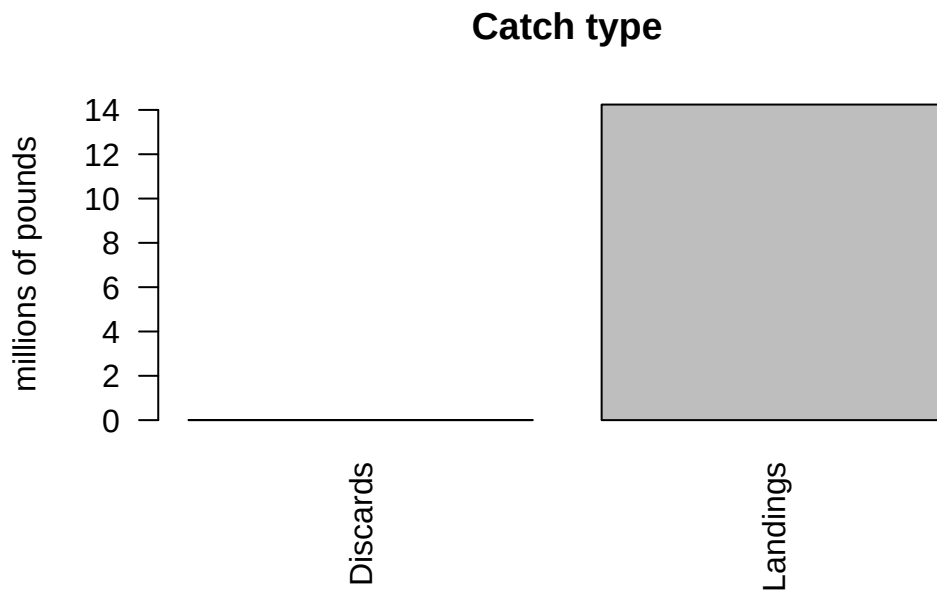


```
barplot(tapply(hs$pounds/106, hs$fishing_sector, sum, na.rm = T), las = 2, ylab = "million
```



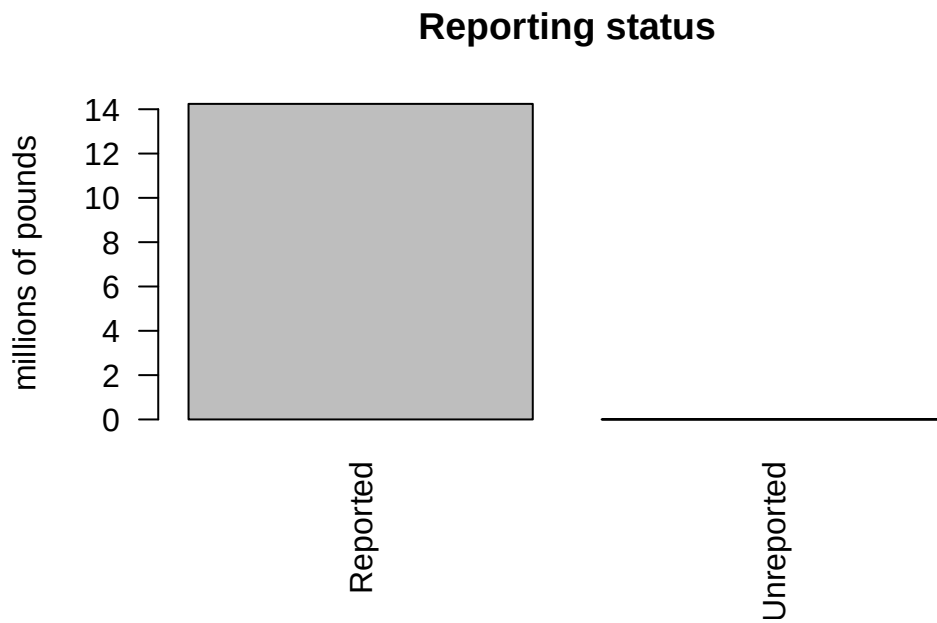
4 Analysis of international landings estimates from Sea Around Us database

```
barplot(tapply(hs$pounds/10^6, hs$catch_type, sum, na.rm = T), las = 2, ylab = "millions of
```

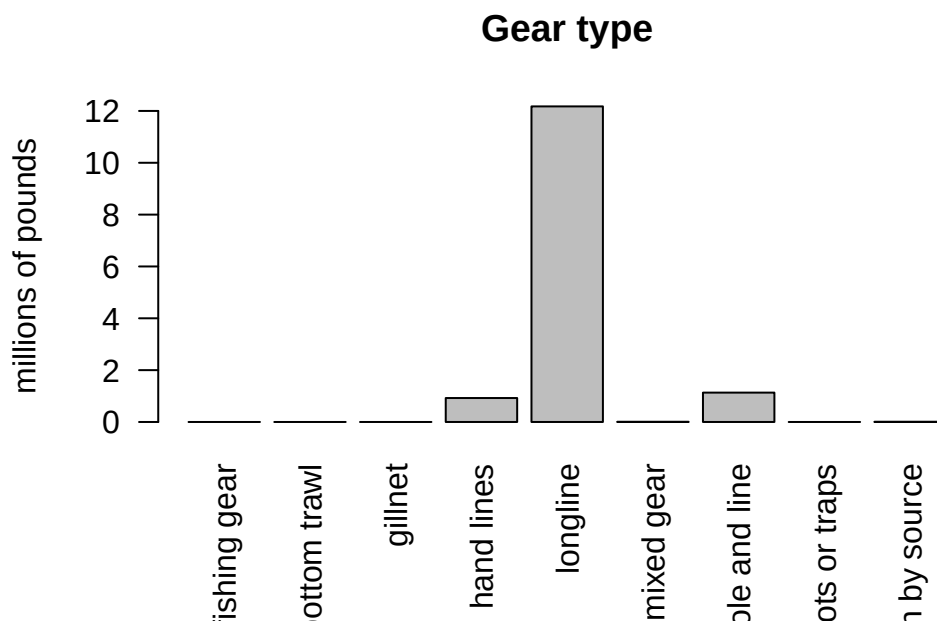


```
barplot(tapply(hs$pounds/10^6, hs$reporting_status, sum, na.rm = T), las = 2, ylab = "milli
```

4 Analysis of international landings estimates from Sea Around Us database



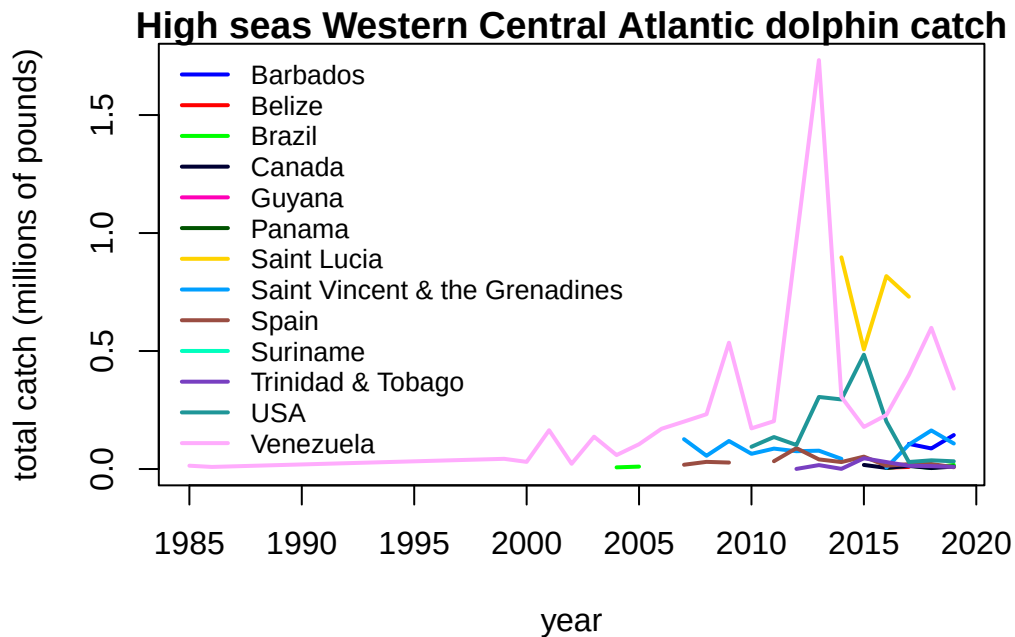
```
barplot(tapply(hs$pounds/106, hs$gear_type, sum, na.rm = T), las = 2, ylab = "millions of pounds")
```



4.3 Summarize the high seas landings

We plot the annual catches by fishing entity to look at trends over time.

```
par(mar = c(5, 5, 1, 1), mfrow = c(1, 1))
tab <- tapply(hs$pounds, list(hs$year, hs$fishing_entity), sum, na.rm = T)
matplot(as.numeric(rownames(tab)), tab/10^6, type = "l", col = glasbey(ncol(tab)), lty = 1,
        main = "High seas Western Central Atlantic dolphin catch")
legend("topleft", colnames(tab), col = glasbey(ncol(tab)), lty = 1, lwd = 2, cex = 0.8, bty
```



Finally we output the subset of high seas dolphin catch for the two areas. Because we have already accounted for the USA landings in our national reporting and are aiming to only summarize international landings here, we filter out the USA landings from the Sea Around Us database.

```
hs <- hs[which(hs$fishing_entity != "USA"), ]
#table(hs$fishing_entity)
hs$year <- factor(hs$year, levels = 1950:2019)
tab <- tapply(hs$pounds, list(hs$year, hs$area_name), sum, na.rm = T)
tab[is.na(tab)] <- 0
tab
```

4 Analysis of international landings estimates from Sea Around Us database

	Atlantic, Northwest	Atlantic, Western Central
1950	0.000	0.000
1951	0.000	0.000
1952	0.000	0.000
1953	0.000	0.000
1954	0.000	0.000
1955	0.000	0.000
1956	0.000	0.000
1957	0.000	0.000
1958	0.000	0.000
1959	0.000	0.000
1960	0.000	0.000
1961	0.000	0.000
1962	0.000	0.000
1963	0.000	0.000
1964	0.000	0.000
1965	0.000	0.000
1966	0.000	0.000
1967	0.000	0.000
1968	0.000	0.000
1969	0.000	0.000
1970	0.000	0.000
1971	0.000	0.000
1972	0.000	0.000
1973	0.000	0.000
1974	0.000	0.000
1975	0.000	0.000
1976	0.000	0.000
1977	0.000	0.000
1978	0.000	0.000
1979	0.000	0.000
1980	0.000	0.000
1981	0.000	0.000
1982	0.000	0.000
1983	0.000	0.000
1984	0.000	0.000
1985	0.000	14037.129
1986	0.000	8748.399
1987	0.000	0.000
1988	0.000	0.000
1989	0.000	0.000

4 Analysis of international landings estimates from Sea Around Us database

1990	0.000	0.000
1991	0.000	0.000
1992	0.000	0.000
1993	0.000	0.000
1994	0.000	0.000
1995	0.000	0.000
1996	0.000	0.000
1997	0.000	0.000
1998	0.000	0.000
1999	0.000	43168.296
2000	0.000	30284.627
2001	0.000	164272.223
2002	0.000	22147.477
2003	0.000	137103.181
2004	0.000	66058.258
2005	0.000	115171.863
2006	0.000	170522.101
2007	14687.527	330838.620
2008	28552.636	340863.719
2009	18719.121	662720.661
2010	4510.085	232093.837
2011	21589.424	300498.224
2012	72398.812	1059210.392
2013	34741.604	2471791.396
2014	27922.953	1247462.983
2015	90313.638	710059.125
2016	15319.433	1094893.891
2017	34806.933	1374259.739
2018	29328.358	885814.754
2019	21870.621	632747.157

```
# write summary to merge with EEZ data
write.csv(tab, file = "data/SAU/high_seas_by_year.csv")
```

4.4 Read in the landings within exclusive economic zones (EEZs)

Now we read in the landings reported by each country or jurisdiction's respective EEZ. Because we have extracted dolphin landings for EEZs worldwide, we need to separate out the landings for the Atlantic Ocean basin.

4 Analysis of international landings estimates from Sea Around Us database

```
rm(list = ls())
d <- read.csv("data/SAU/SAU Taxa 600006 v50-1.csv", header = T, sep = ",") # v. 50.1 - Jun
head(d)
```

	area_name	area_type	year	scientific_name	common_name
1	Australia	eez	1950	Coryphaena hippurus	Common dolphinfish
2	Australia	eez	1950	Coryphaena hippurus	Common dolphinfish
3	Australia	eez	1950	Coryphaena hippurus	Common dolphinfish
4	Bahamas	eez	1950	Coryphaena hippurus	Common dolphinfish
5	Barbados	eez	1950	Coryphaena hippurus	Common dolphinfish
6	Bermuda (UK)	eez	1950	Coryphaena hippurus	Common dolphinfish

	functional_group	commercial_group	fishing_entity	fishing_sector
1	Large pelagics (>=90 cm)	Perch-likes	Australia	Industrial
2	Large pelagics (>=90 cm)	Perch-likes	Australia	Industrial
3	Large pelagics (>=90 cm)	Perch-likes	Australia	Industrial
4	Large pelagics (>=90 cm)	Perch-likes	Bahamas	Recreational
5	Large pelagics (>=90 cm)	Perch-likes	Barbados	Recreational
6	Large pelagics (>=90 cm)	Perch-likes	Bermuda (UK)	Artisanal

	catch_type	reporting_status	gear_type
1	Landings	Reported	longline
2	Landings	Reported	longline
3	Landings	Reported	longline
4	Landings	Unreported	recreational fishing gear
5	Landings	Unreported	recreational fishing gear
6	Landings	Reported	small scale lines

	end_use_type	tonnes	landed_value
1	Fishmeal and fish oil	0.003373202	3.438170e-05
2	Direct human consumption	6.739657104	1.040254e+04
3	Other	0.003373202	3.438170e-05
4	Direct human consumption	48.665105143	8.262728e+00
5	Direct human consumption	0.009158559	4.846886e-04
6	Direct human consumption	1.511085514	3.199403e+03

```
#dim(d)
apply(d[1:13], 2, table)
```

\$area_name

Albania

4 Analysis of international landings estimates from Sea Around Us database

	1
Algeria	
	70
All	
	312
American Samoa	
	262
Andaman & Nicobar Isl. (India)	
	883
Angola	
	78
Anguilla (UK)	
	5
Antigua & Barbuda	
	6
Aruba (Netherlands)	
	470
Ascension Isl. (UK)	
	1
Australia	
	786
Azores Isl. (Portugal)	
	303
Bahamas	
	135
Balearic Islands (Spain)	
	2147
Bangladesh	
	5
Barbados	
	87
Belize	
	11
Benin	
	121
Bermuda (UK)	
	136
Bonaire (Netherlands)	
	137
Brazil (mainland)	
	1286

4 Analysis of international landings estimates from Sea Around Us database

British Virgin Isl. (UK)	4
Brunei Darussalam	84
Cambodia	5
Cameroon	10
Canada (East Coast)	135
Canary Isl. (Spain)	290
Cape Verde	702
Cayman Isl. (UK)	277
Chagos Archipelago (UK)	6
China	2494
Christmas Isl. (Australia)	90
Clipperton Isl. (France)	225
Colombia (Caribbean)	152
Colombia (Pacific)	171
Comoros Isl.	83
Congo (ex-Zaire)	5
Congo, R. of	13
Cook Islands	368
Corsica (France)	21
Costa Rica (Caribbean)	38
Costa Rica (Pacific)	

4 Analysis of international landings estimates from Sea Around Us database

	180
Côte d'Ivoire	
	157
Crete (Greece)	
	6
Croatia	
	9
Cuba	
	2
Curaçao (Netherlands)	
	142
Cyprus (South)	
	38
Desventuradas Isl. (Chile)	
	1
Dominica	
	297
Dominican Republic	
	298
Easter Isl. (Chile)	
	96
Ecuador (mainland)	
	142
El Salvador	
	83
Equatorial Guinea	
	297
Fernando de Noronha (Brazil)	
	446
Fiji	
	386
France (Atlantic Coast)	
	216
France (Mediterranean)	
	57
French Guiana	
	42
French Polynesia	
	332
Gabon	
	158

4 Analysis of international landings estimates from Sea Around Us database

Galapagos Isl. (Ecuador)	
	237
Gambia	
	125
Ghana	
	105
Glorieuse Islands (France)	
	6
Greece (without Crete)	
	17
Grenada	
	373
Guadeloupe (France)	
	132
Guam (USA)	
	91
Guatemala (Caribbean)	
	1
Guatemala (Pacific)	
	390
Guinea	
	166
Guinea-Bissau	
	146
Guyana	
	55
Haiti	
	316
Hawaii Main Islands (USA)	
	665
Hawaii Northwest Islands (USA)	
	523
Honduras (Caribbean)	
	373
Honduras (Pacific)	
	21
Hong Kong (China)	
	288
Howland & Baker Isl. (USA)	
	172
India (mainland)	

4 Analysis of international landings estimates from Sea Around Us database

	33
Indonesia (Central)	
	154
Indonesia (Eastern)	
	431
Indonesia (Indian Ocean)	
	169
Iran (Sea of Oman)	
	283
Italy (mainland)	
	1662
Jamaica	
	135
Japan (Daito Islands)	
	337
Japan (main islands)	
	386
Japan (Ogasawara Islands)	
	334
Jarvis Isl. (USA)	
	107
Johnston Atoll (USA)	
	171
Kenya	
	68
Kermadec Isl. (New Zealand)	
	187
Kiribati (Gilbert Islands)	
	519
Kiribati (Line Islands)	
	505
Kiribati (Phoenix Islands)	
	526
Korea (North, Sea of Japan)	
	162
Korea (North, Yellow Sea)	
	162
Korea (South)	
	347
Lebanon	
	12

4 Analysis of international landings estimates from Sea Around Us database

Liberia	
	183
Libya	
	818
Lord Howe Isl. (Australia)	
	63
Madagascar	
	199
Madeira Isl. (Portugal)	
	135
Malaysia (Peninsula East)	
	7
Malaysia (Peninsula West)	
	49
Malaysia (Sabah)	
	71
Malaysia (Sarawak)	
	65
Maldives	
	22
Malta	
	1946
Marshall Isl.	
	504
Martinique (France)	
	123
Mauritania	
	183
Mauritius	
	236
Mayotte (France)	
	365
Mexico (Atlantic)	
	319
Mexico (Pacific)	
	759
Micronesia (Federated States of)	
	572
Montserrat (UK)	
	250
Morocco (Central)	

4 Analysis of international landings estimates from Sea Around Us database

	147
Morocco (Mediterranean)	
	108
Morocco (South)	
	159
Mozambique	
	75
Mozambique Channel Isl. (France)	
	32
Myanmar	
	48
Namibia	
	27
Nauru	
	367
New Caledonia (France)	
	170
New Zealand	
	1276
Nicaragua (Caribbean)	
	3
Nicaragua (Pacific)	
	126
Nigeria	
	31
Niue (New Zealand)	
	310
Norfolk Isl. (Australia)	
	56
Northern Marianas (USA)	
	446
Oman	
	14
Pakistan	
	1898
Palau	
	427
Palmyra Atoll & Kingman Reef (USA)	
	101
Panama (Caribbean)	
	34

4 Analysis of international landings estimates from Sea Around Us database

Panama (Pacific)	179
Papua New Guinea	615
Peru	1312
Philippines	1124
Pitcairn (UK)	181
Portugal (mainland)	80
Puerto Rico (USA)	359
Réunion (France)	106
Saba and Sint Eustatius (Netherlands)	84
Saint Helena (UK)	22
Saint Kitts & Nevis	324
Saint Lucia	379
Saint Pierre & Miquelon (France)	3
Saint Vincent & the Grenadines	466
Samoa	456
Sao Tome & Principe	472
Sardinia (Italy)	1634
Senegal	218
Seychelles	205
Sicily (Italy)	1657
Sierra Leone	

4 Analysis of international landings estimates from Sea Around Us database

	158
Singapore	
	22
Sint Maarten (Netherlands)	
	106
Solomon Isl.	
	515
Somalia	
	12
South Africa (Atlantic and Cape)	
	216
South Africa (Indian Ocean Coast)	
	298
Spain (mainland, Med and Gulf of Cadiz)	
	860
Spain (Northwest)	
	142
Sri Lanka	
	18
St Barthelemy (France)	
	429
St Martin (France)	
	429
St Paul & Amsterdam Isl. (France)	
	3
St Paul and St. Peter Archipelago (Brazil)	
	250
Suriname	
	53
Syria	
	329
Taiwan	
	1379
Tanzania	
	58
Thailand (Andaman Sea)	
	41
Thailand (Gulf of Thailand)	
	41
Timor Leste	
	46

4 Analysis of international landings estimates from Sea Around Us database

Togo	
	247
Tokelau (New Zealand)	
	445
Tonga	
	360
Trindade & Martim Vaz Isl. (Brazil)	
	64
Trinidad & Tobago	
	161
Tristan da Cunha Isl. (UK)	
	1
Tromelin Isl. (France)	
	15
Tunisia	
	649
Turks & Caicos Isl. (UK)	
	225
Tuvalu	
	452
Uruguay	
	3
US Virgin Isl.	
	142
USA (East Coast)	
	2668
USA (Gulf of Mexico)	
	1374
USA (West Coast)	
	365
Vanuatu	
	384
Venezuela	
	120
Viet Nam	
	335
Wake Isl. (USA)	
	261
Wallis & Futuna Isl. (France)	
	398
Yemen (Socotra)	

4 Analysis of international landings estimates from Sea Around Us database

2

\$area_type

eez high_seas
63056 312

\$year

1950	1951	1952	1953	1954	1955	1956	1957	1958	1959	1960	1961	1962	1963	1964	1965
436	453	477	472	500	507	508	514	514	543	543	545	530	532	600	635
1966	1967	1968	1969	1970	1971	1972	1973	1974	1975	1976	1977	1978	1979	1980	1981
670	666	671	698	719	760	743	738	799	775	791	691	717	692	694	703
1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
736	688	684	746	739	791	811	827	864	930	927	907	917	940	911	952
1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
972	1002	1015	1053	1214	1225	1290	1250	1245	1318	1300	1269	1372	1457	1401	1584
2014	2015	2016	2017	2018	2019										
1687	1647	1680	1703	1715	1763										

\$scientific_name

Coryphaena hippurus
63368

\$common_name

Common dolphinfish
63368

\$functional_group

Large pelagics (≥ 90 cm)
63368

\$commercial_group

Perch-likes
63368

\$fishing_entity

4 Analysis of international landings estimates from Sea Around Us database

Algeria	American Samoa
40	157
Aruba (Netherlands)	Australia
45	455
Azores Isl. (Portugal)	Bahamas
212	70
Barbados	Belize
76	14
Benin	Bermuda (UK)
72	134
Bonaire (Netherlands)	Brazil
70	2054
Canada	Cape Verde
132	624
Cayman Isl. (UK)	Channel Isl. (UK)
356	1
Chile	China
18	2500
Christmas Isl. (Australia)	Colombia
28	24
Comoros	Costa Rica
36	64
Côte d'Ivoire	Cuba
11	255
Curacao	Dominica
71	330
Dominican Republic	Ecuador
293	114
Equatorial Guinea	Fiji
260	336
France	French Polynesia
776	137
Ghana	Grenada
46	462
Guadeloupe (France)	Guam (USA)
129	33
Guatemala	Guyana
313	3
Haiti	Honduras
237	287

4 Analysis of international landings estimates from Sea Around Us database

India	Indonesia
875	255
Iran	Italy
280	4883
Jamaica	Japan
123	3322
Kenya	Korea (South)
55	2308
Lebanon	Liberia
12	41
Libya	Madagascar
720	71
Madeira Isl. (Portugal)	Maldives
90	7
Malta	Marshall Isl.
2190	303
Martinique (France)	Mauritania
355	64
Mauritius	Mayotte (France)
104	366
Mexico	Montserrat (UK)
1016	243
Morocco	Mozambique
75	11
New Zealand	Nicaragua
424	52
Niue (New Zealand)	North Marianas (USA)
154	155
Oman	Pakistan
14	1876
Palau	Panama
16	120
Papua New Guinea	Peru
390	1100
Philippines	Portugal
934	658
Puerto Rico (USA)	Réunion (France)
309	176
Saba and Saint Eustaius (Netherlands)	Saint Helena (UK)
70	22
Saint Kitts & Nevis	Saint Lucia

4 Analysis of international landings estimates from Sea Around Us database

321	234
Saint Vincent & the Grenadines	Samoa
306	157
Sao Tome & Principe	Senegal
342	107
Seychelles	Sint Maarten
136	70
Solomon Isl.	South Africa
114	453
South Cyprus	Spain
38	5112
Sri Lanka	St Barthelemy (France)
26	418
St Martin	Suriname
418	11
Syrian Arab Republic	Taiwan
329	6240
Tanzania	Thailand
68	30
Togo	Tokelau (New Zealand)
129	157
Tonga	Trinidad & Tobago
115	103
Tunisia	Turks & Caicos Isl. (UK)
551	223
Ukraine	United Kingdom
162	10
Unknown Fishing Country	US Virgin Isl.
3364	133
USA	Vanuatu
6780	486
Venezuela	Viet Nam
470	75
Wallis & Futuna Isl. (France)	
166	

\$fishing_sector

Artisanal	Industrial	Recreational	Subsistence
25616	33126	2637	1989

4 Analysis of international landings estimates from Sea Around Us database

\$catch_type

Discards	Landings
17484	45884

\$reporting_status

Reported	Unreported
25295	38073

\$gear_type

artisanal fishing gear	bagnets
1807	28
bottom trawl	cast nets
2101	28
dragged gear	dredge
12	45
encircling nets	gillnet
12	1083
hand lines	hand or tools
1172	117
harpoon	lines
57	1984
longline	mixed gear
13940	500
other	other industrial
208	335
other nets	otter trawl
122	6
pelagic trawl	pole and line
397	577
pots or traps	purse seine
269	9699
recreational fishing gear	shrimp trawl
2358	205
small encircling nets	small scale encircling nets
15	2517
small scale gillnets	small scale hand lines
4701	669
small scale lines	small scale longline

4 Analysis of international landings estimates from Sea Around Us database

	7027		551
small scale other nets		small scale pole lines	
	531		453
small scale pots or traps		small scale purse seine	
	2441		261
small scale seine nets		small scale trammel net	
	1338		41
subsistence fishing gear		unknown by author	
	1831		97
unknown by source		unknown class	
	312		3521

\$end_use_type

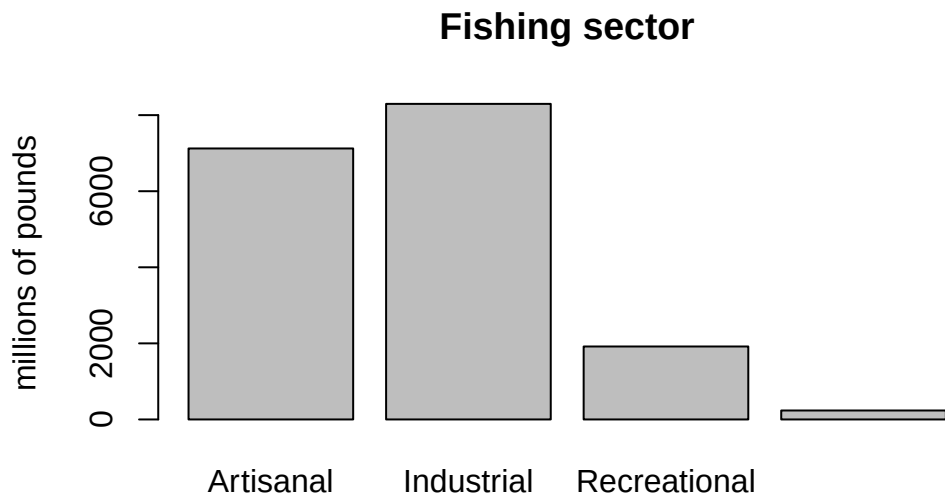
	Direct human consumption	Fishmeal and fish oil
17484	25810	8652
Other		
11422		

```
# convert tonnes to pounds
d$lbs <- d$tonnes * 1000 * 2.20462
```

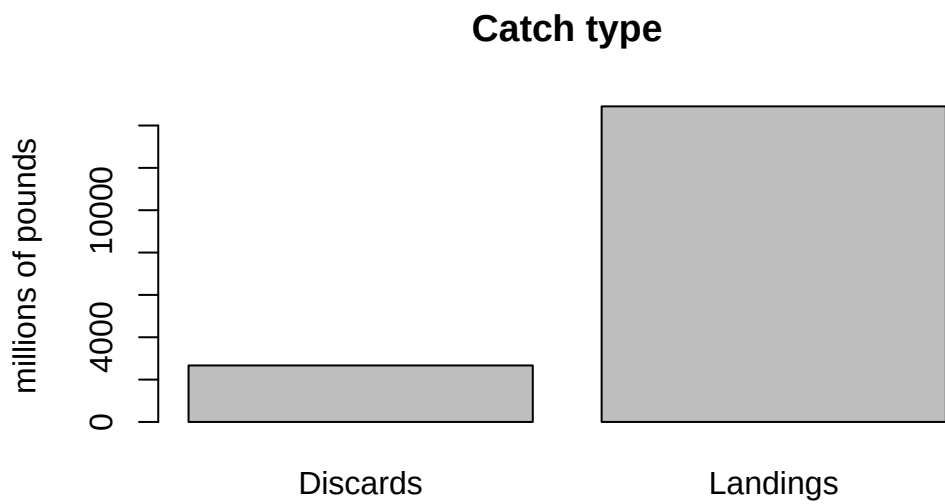
We look at the characteristics of the landings within EEZx. These are largely industrial and artisanal in nature. Landings are the majority of the catch but discards are also present. There are more unreported landings than reported landings and a variety of gear types are used.

```
# look at characteristics of data
#par(mfrow = c(2, 2), mar = c(5, 5, 3, 1))
barplot(tapply(d$lbs, d$fishing_sector, sum, na.rm = T)/10^6, ylab = "millions of pounds",
```

4 Analysis of international landings estimates from Sea Around Us database

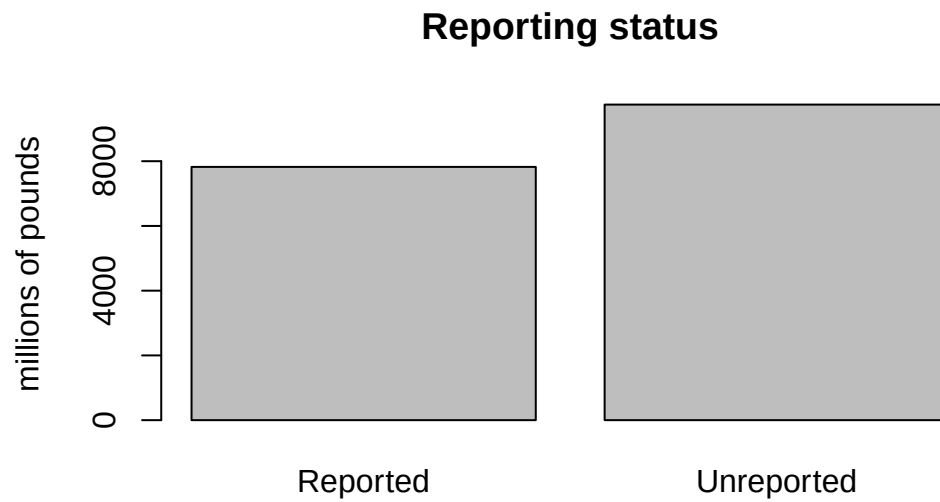


```
barplot(tapply(d$lbs, d$catch_type, sum, na.rm = T)/10^6, ylab = "millions of pounds", main = "Catch type")
```



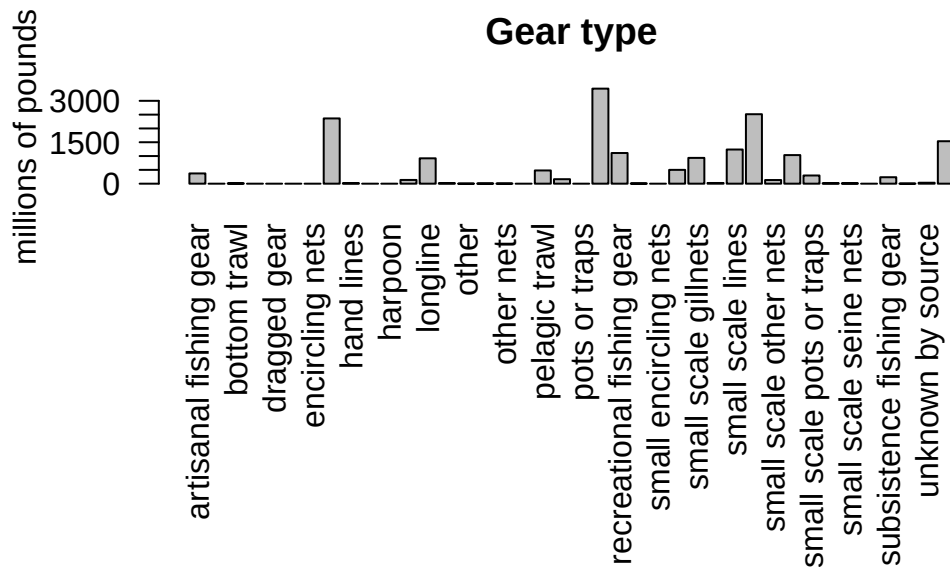
4 Analysis of international landings estimates from Sea Around Us database

```
barplot(tapply(d$lbs, d$reporting_status, sum, na.rm = T)/106, ylab = "millions of pounds")
```



```
par(mar= c(12, 5, 3, 1))  
barplot(tapply(d$lbs, d$gear_type, sum, na.rm = T)/106, las = 2, ylab = "millions of pounds")
```

4 Analysis of international landings estimates from Sea Around Us database



4.5 Separating the EEZs by area

Now we have to manually assign the different EEZs to respective areas. We separate the geographical areas within the Atlantic Ocean (North and South) and lump all other EEZs into “other oceans.” This latter category is largely the Pacific Ocean and to a lesser extent the Indian Ocean. We remove the high seas catch from this database as it is not ocean-specific and we have extracted the ocean-specific high seas catch previously.

```
# remove the high seas catch from this database
d <- d[which(d$area_type != "high_seas"), ]
d$fishing_entity <- as.factor(d$fishing_entity)
d$fishing_entity <- droplevels(d$fishing_entity)
table(d$area_type)
```

```
eez
63056
```

4 Analysis of international landings estimates from Sea Around Us database

```
# Western Central Atlantic EEZs
WCA <- c("Bahamas", "Barbados", "Cayman Isl. (UK)", "Dominica", "Dominican Republic", "Grenada",
        "Aruba (Netherlands)", "Haiti", "Guadeloupe (France)", "Jamaica", "Martinique (France)",
        "Montserrat (UK)", "Nicaragua (Caribbean)", "Puerto Rico (USA)", "Saint Kitts & Nevis",
        "Saint Lucia", "Saint Vincent & the Grenadines", "Trinidad & Tobago", "US Virgin Islands",
        "St Barthelemy (France)", "St Martin (France)", "Curaçao (Netherlands)", "Bonaire",
        "Saba and Sint Eustatius (Netherlands)", "Sint Maarten (Netherlands)", "Honduras (Central America)",
        "Colombia (Caribbean)", "Mexico (Atlantic)", "Turks & Caicos Isl. (UK)", "Guyana",
        "Antigua & Barbuda", "Belize", "British Virgin Isl. (UK)", "French Guiana", "Cuba",
        "Anguilla (UK)", "Suriname", "Costa Rica (Caribbean)", "Guatemala (Caribbean)",
        "Panama (Caribbean)", "Bermuda (UK)", "Curaçao (Netherlands)")

# Brazil
Bra <- c("Brazil (mainland)", "Fernando de Noronha (Brazil)",
        "St Paul and St. Peter Archipelago (Brazil)")

# Canada
Can <- c("Canada (East Coast)", "Saint Pierre & Miquelon (France)")

# United States
USA <- c("USA (East Coast)", "USA (Gulf of Mexico)")

# Northeast Atlantic EEZs
NEA <- c("Canary Isl. (Spain)", "Spain (mainland; Med and Gulf of Cadiz)", "Spain (Northwest Africa)",
        "Spain (mainland, Med and Gulf of Cadiz)",
        "France (Atlantic Coast)", "Portugal (mainland)", "Azores Isl. (Portugal)")

# Eastern Central Atlantic EEZs
ECA <- c("Madeira Isl. (Portugal)", "Cape Verde", "Equatorial Guinea", "Benin",
        "Sao Tome & Principe ", "Sao Tome & Principe", "Morocco (Central)", "Ghana",
        "Côte d'Ivoire", "Côte d'Ivoire", "Guinea", "Liberia", "Sierra Leone", "Nigeria",
        "Guinea-Bissau", "Senegal", "Cameroon", "Gambia", "Mauritania", "Morocco (South)",
        "Gabon", "Congo; R. of", "Congo, R. of", "Congo (ex-Zaire)")

# Southeastern Atlantic EEZs
SEA <- c("Angola", "Namibia", "South Africa (Atlantic and Cape)",
        "Ascension Isl. (UK)", "Saint Helena (UK)", "Tristan da Cunha Isl. (UK)")

# Southwest Atlantic EEZs
SWA <- c("Trindade & Martim Vaz Isl. (Brazil)", "Uruguay")
```

4 Analysis of international landings estimates from Sea Around Us database

```
# Mediterranean EEZs
Med <- c("Italy (mainland)", "Libya", "Malta", "Syria", "Sicily (Italy)", "Sardinia (Italy)",
        "Balearic Islands (Spain)", "Cyprus (North)", "Cyprus (South)",
        "Gaza Strip", "Greece (without Crete)", "Crete (Greece)", "Lebanon",
        "Turkey (Mediterranean Sea)", "Egypt (Mediterranean)", "Israel (Mediterranean)",
        "Algeria", "Albania", "Croatia", "Montenegro", "Morocco (Mediterranean)",
        "France (Mediterranean)", "Corsica (France)")

# add variable to specify region
d$reg <- "other oceans"
d$reg[which(d$area_name %in% WCA)] <- "Western Central EEZs"
d$reg[which(d$area_name %in% Bra)] <- "Brazil EEZ"
d$reg[which(d$area_name %in% Can)] <- "Canadian EEZ"
d$reg[which(d$area_name %in% USA)] <- "United States EEZ"
d$reg[which(d$area_name %in% NEA)] <- "Northeast EEZs"
d$reg[which(d$area_name %in% ECA)] <- "Eastern Central EEZs"
d$reg[which(d$area_name %in% SEA)] <- "Southeast EEZs"
d$reg[which(d$area_name %in% SWA)] <- "Southwest EEZs"
d$reg[which(d$area_name %in% Med)] <- "Mediterranean Sea EEZs"
```

Now we will check that we have classified the regions correctly.

```
# check region designation
othlis <- unique(d$area_name[which(d$reg == "other oceans")])
print("EEZs of all other oceans: ")
```

```
[1] "EEZs of all other oceans: "
```

```
othlis
```

[1] "Australia"	"China"
[3] "Taiwan"	"Mayotte (France)"
[5] "French Polynesia"	"Hong Kong (China)"
[7] "Andaman & Nicobar Isl. (India)"	"Indonesia (Eastern)"
[9] "Japan (main islands)"	"Japan (Daito Islands)"
[11] "Korea (South)"	"Hawaii Northwest Islands (USA)"
[13] "Nauru"	"Niue (New Zealand)"
[15] "Northern Marianas (USA)"	"Micronesia (Federated States of)"

4 Analysis of international landings estimates from Sea Around Us database

[17] "Marshall Isl."	"Palau"
[19] "Pakistan"	"Papua New Guinea"
[21] "Philippines"	"Seychelles"
[23] "Viet Nam"	"Tokelau (New Zealand)"
[25] "Tonga"	"Hawaii Main Islands (USA)"
[27] "USA (West Coast)"	"Wake Isl. (USA)"
[29] "Wallis & Futuna Isl. (France)"	"Samoa"
[31] "Guatemala (Pacific)"	"Kiribati (Gilbert Islands)"
[33] "Mexico (Pacific)"	"Japan (Ogasawara Islands)"
[35] "Korea (North, Yellow Sea)"	"Korea (North, Sea of Japan)"
[37] "Mauritius"	"Solomon Isl."
[39] "Christmas Isl. (Australia)"	"Johnston Atoll (USA)"
[41] "Timor Leste"	"Palmyra Atoll & Kingman Reef (USA)"
[43] "Jarvis Isl. (USA)"	"Howland & Baker Isl. (USA)"
[45] "Indonesia (Central)"	"Indonesia (Indian Ocean)"
[47] "Kiribati (Line Islands)"	"Kiribati (Phoenix Islands)"
[49] "New Caledonia (France)"	"Vanuatu"
[51] "Lord Howe Isl. (Australia)"	"Myanmar"
[53] "Ecuador (mainland)"	"El Salvador"
[55] "Fiji"	"Malaysia (Peninsula West)"
[57] "Norfolk Isl. (Australia)"	"Peru"
[59] "Pitcairn (UK)"	"Colombia (Pacific)"
[61] "Costa Rica (Pacific)"	"Nicaragua (Pacific)"
[63] "Panama (Pacific)"	"Thailand (Andaman Sea)"
[65] "Thailand (Gulf of Thailand)"	"Cook Islands"
[67] "Kermadec Isl. (New Zealand)"	"Tuvalu"
[69] "Honduras (Pacific)"	"American Samoa"
[71] "Galapagos Isl. (Ecuador)"	"New Zealand"
[73] "Brunei Darussalam"	"Malaysia (Sabah)"
[75] "Malaysia (Sarawak)"	"Guam (USA)"
[77] "Clipperton Isl. (France)"	"South Africa (Indian Ocean Coast)"
[79] "Singapore"	"Easter Isl. (Chile)"
[81] "Madagascar"	"Réunion (France)"
[83] "Chagos Archipelago (UK)"	"Comoros Isl."
[85] "Mozambique Channel Isl. (France)"	"Kenya"
[87] "Mozambique"	"Somalia"
[89] "Tanzania"	"Glorieuse Islands (France)"
[91] "Yemen (Socotra)"	"Oman"
[93] "St Paul & Amsterdam Isl. (France)"	"Iran (Sea of Oman)"
[95] "Desventuradas Isl. (Chile)"	"India (mainland)"
[97] "Maldives"	"Tromelin Isl. (France)"

4 Analysis of international landings estimates from Sea Around Us database

```
[99] "Cambodia"                "Sri Lanka"  
[101] "Malaysia (Peninsula East)" "Bangladesh"
```

```
print("U.S. EEZs outside the Atlantic Ocean: ")
```

```
[1] "U.S. EEZs outside the Atlantic Ocean: "
```

```
othlis[grep("USA", othlis)]
```

```
[1] "Hawaii Northwest Islands (USA)"    "Northern Marianas (USA)"  
[3] "Hawaii Main Islands (USA)"         "USA (West Coast)"  
[5] "Wake Isl. (USA)"                  "Johnston Atoll (USA)"  
[7] "Palmyra Atoll & Kingman Reef (USA)" "Jarvis Isl. (USA)"  
[9] "Howland & Baker Isl. (USA)"        "Guam (USA)"
```

```
print("French EEZs outside the Atlantic Ocean: ")
```

```
[1] "French EEZs outside the Atlantic Ocean: "
```

```
othlis[grep("France", othlis)]
```

```
[1] "Mayotte (France)"                "Wallis & Futuna Isl. (France)"  
[3] "New Caledonia (France)"          "Clipperton Isl. (France)"  
[5] "Réunion (France)"                "Mozambique Channel Isl. (France)"  
[7] "Glorieuse Islands (France)"      "St Paul & Amsterdam Isl. (France)"  
[9] "Tromelin Isl. (France)"
```

```
print("Western Central EEZs:")
```

```
[1] "Western Central EEZs:"
```

```
unique(d$area_name[which(d$reg == "Western Central EEZs")])
```

4 Analysis of international landings estimates from Sea Around Us database

- [1] "Bahamas"
- [2] "Barbados"
- [3] "Bermuda (UK)"
- [4] "Cayman Isl. (UK)"
- [5] "Dominica"
- [6] "Dominican Republic"
- [7] "Grenada"
- [8] "Guadeloupe (France)"
- [9] "Haiti"
- [10] "Jamaica"
- [11] "Martinique (France)"
- [12] "Montserrat (UK)"
- [13] "Puerto Rico (USA)"
- [14] "Saint Kitts & Nevis"
- [15] "Saint Lucia"
- [16] "Saint Vincent & the Grenadines"
- [17] "Trinidad & Tobago"
- [18] "US Virgin Isl."
- [19] "St Barthelemy (France)"
- [20] "St Martin (France)"
- [21] "Curaçao (Netherlands)"
- [22] "Bonaire (Netherlands)"
- [23] "Saba and Sint Eustatius (Netherlands)"
- [24] "Sint Maarten (Netherlands)"
- [25] "Honduras (Caribbean)"
- [26] "Colombia (Caribbean)"
- [27] "Mexico (Atlantic)"
- [28] "Turks & Caicos Isl. (UK)"
- [29] "Costa Rica (Caribbean)"
- [30] "Panama (Caribbean)"
- [31] "Antigua & Barbuda"
- [32] "Belize"
- [33] "British Virgin Isl. (UK)"
- [34] "Cuba"
- [35] "French Guiana"
- [36] "Guyana"
- [37] "Anguilla (UK)"
- [38] "Suriname"
- [39] "Venezuela"
- [40] "Guatemala (Caribbean)"
- [41] "Aruba (Netherlands)"

4 Analysis of international landings estimates from Sea Around Us database

```
[42] "Nicaragua (Caribbean)"
```

```
print("Brazil EEZs:")
```

```
[1] "Brazil EEZs:"
```

```
unique(d$area_name[which(d$reg == "Brazil EEZ")])
```

```
[1] "Brazil (mainland)"
```

```
[2] "Fernando de Noronha (Brazil)"
```

```
[3] "St Paul and St. Peter Archipelago (Brazil)"
```

```
print("Canadian EEZs:")
```

```
[1] "Canadian EEZs:"
```

```
unique(d$area_name[which(d$reg == "Canadian EEZ")])
```

```
[1] "Canada (East Coast)"
```

```
"Saint Pierre & Miquelon (France)"
```

```
print("United States EEZs:")
```

```
[1] "United States EEZs:"
```

```
unique(d$area_name[which(d$reg == "United States EEZ")])
```

```
[1] "USA (East Coast)"
```

```
"USA (Gulf of Mexico)"
```

```
print("Northeast Atlantic EEZs:")
```

```
[1] "Northeast Atlantic EEZs:"
```

4 Analysis of international landings estimates from Sea Around Us database

```
unique(d$area_name[which(d$reg == "Northeast EEZs")])
```

```
[1] "Azores Isl. (Portugal)"
[2] "Spain (mainland, Med and Gulf of Cadiz)"
[3] "Canary Isl. (Spain)"
[4] "Portugal (mainland)"
[5] "France (Atlantic Coast)"
[6] "Spain (Northwest)"
```

```
print("Eastern Central Atlantic EEZs:")
```

```
[1] "Eastern Central Atlantic EEZs:"
```

```
unique(d$area_name[which(d$reg == "Eastern Central EEZs")])
```

```
[1] "Cape Verde"           "Equatorial Guinea"
[3] "Sao Tome & Principe"  "Madeira Isl. (Portugal)"
[5] "Togo"                 "Ghana"
[7] "Guinea"               "Liberia"
[9] "Guinea-Bissau"        "Senegal"
[11] "Sierra Leone"        "Morocco (Central)"
[13] "Morocco (South)"      "Côte d'Ivoire"
[15] "Gabon"                "Benin"
[17] "Gambia"               "Mauritania"
[19] "Nigeria"              "Cameroon"
[21] "Congo, R. of"         "Congo (ex-Zaire)"
```

```
print("Southeast EEZs:")
```

```
[1] "Southeast EEZs:"
```

```
unique(d$area_name[which(d$reg == "Southeast EEZs")])
```

```
[1] "Ascension Isl. (UK)"      "South Africa (Atlantic and Cape)"
[3] "Saint Helena (UK)"       "Angola"
[5] "Namibia"                 "Tristan da Cunha Isl. (UK)"
```

4 Analysis of international landings estimates from Sea Around Us database

```
print("Southwest EEZs:")
```

```
[1] "Southwest EEZs:"
```

```
unique(d$area_name[which(d$reg == "Southwest EEZs")])
```

```
[1] "Trindade & Martim Vaz Isl. (Brazil)" "Uruguay"
```

```
print("Mediterranean Sea EEZs:")
```

```
[1] "Mediterranean Sea EEZs:"
```

```
unique(d$area_name[which(d$reg == "Mediterranean Sea EEZs")])
```

```
[1] "Italy (mainland)"      "Libya"
[3] "Malta"                 "Syria"
[5] "Sicily (Italy)"        "Sardinia (Italy)"
[7] "Balearic Islands (Spain)" "Tunisia"
[9] "Algeria"               "Morocco (Mediterranean)"
[11] "France (Mediterranean)" "Corsica (France)"
[13] "Croatia"               "Lebanon"
[15] "Albania"               "Greece (without Crete)"
[17] "Crete (Greece)"        "Cyprus (South)"
```

4.6 Summarize catch by regional EEZs

Now we will summarize the catch by region by grouping the catch across the EEZs by region and plot the results.

```
# summarize EEZ catch by region
tab <- tapply(d$lbs, list(d$year, d$reg), sum, na.rm = T)
head(tab)
```

4 Analysis of international landings estimates from Sea Around Us database

	Brazil EEZ	Canadian EEZ	Eastern Central EEZs	Mediterranean Sea EEZs
1950	1127242	NA	677552.5	26162262
1951	1060734	NA	693004.4	25410830
1952	1128992	NA	708466.7	30206350
1953	1111008	NA	723939.4	31275619
1954	1090492	NA	739422.5	31383840
1955	1190424	NA	754915.9	30056210
	Northeast EEZs	other oceans	Southeast EEZs	Southwest EEZs
1950	1173441	19331387	NA	NA
1951	1187386	21573930	NA	NA
1952	1193045	22569230	NA	NA
1953	1141504	25904815	NA	NA
1954	1269523	32505789	NA	NA
1955	1405543	34571430	NA	NA
	United States EEZ	Western Central EEZs		
1950	1548245	1345027		
1951	1721403	1467925		
1952	1871779	1510406		
1953	2024535	2001465		
1954	2177865	2069754		
1955	2309815	2130341		

```

tab2 <- t(tab)
tab3 <- tab2[order(rowSums(tab2, na.rm = T), decreasing = T), ]
tab_glob <- t(tab3)
head(tab_glob)

```

	other oceans	Mediterranean Sea EEZs	United States EEZ	Brazil EEZ
1950	19331387	26162262	1548245	1127242
1951	21573930	25410830	1721403	1060734
1952	22569230	30206350	1871779	1128992
1953	25904815	31275619	2024535	1111008
1954	32505789	31383840	2177865	1090492
1955	34571430	30056210	2309815	1190424
	Western Central EEZs	Northeast EEZs	Eastern Central EEZs	Southwest EEZs
1950	1345027	1173441	677552.5	NA
1951	1467925	1187386	693004.4	NA
1952	1510406	1193045	708466.7	NA
1953	2001465	1141504	723939.4	NA
1954	2069754	1269523	739422.5	NA

4 Analysis of international landings estimates from Sea Around Us database

1955	2130341	1405543	754915.9	NA
	Southeast EEZs	Canadian EEZ		
1950	NA	NA		
1951	NA	NA		
1952	NA	NA		
1953	NA	NA		
1954	NA	NA		
1955	NA	NA		

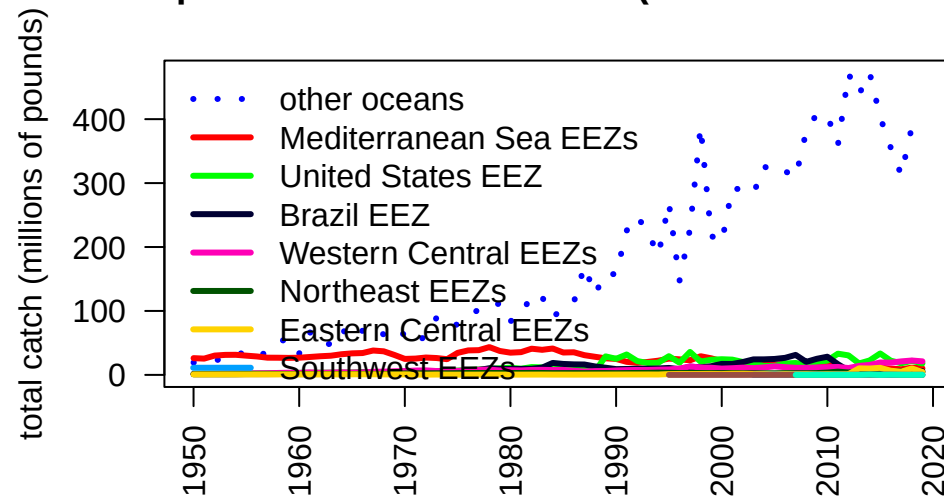
```

yrs <- as.numeric(rownames(tab))

par(mar = c(5, 4, 4, 3), mgp = c(3, 1, 0))
matplot(yrs, tab_glob/10^6, las = 2, type = "l", lty = c(3, rep(1, 9)), lwd = 3, col = glasbey(ncol(tab_glob)),
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Global dolphin catches within EEZs (commercial + recreational)")
legend("topleft", colnames(tab_glob), col = glasbey(ncol(tab_glob)),
        lty = c(3, rep(1, 9)), lwd = 3, bty = "n")

```

Global dolphin catches within EEZs (commercial + recreational)



We can see that the EEZ catches globally are dominated by areas outside of the Atlantic ("other oceans") so we will replot the data after moving this category. Here we can see that most of the catches historically were coming from the Mediterranean Sea, though

4 Analysis of international landings estimates from Sea Around Us database

these catches have been declining drastically since the 1970s. Catches in the United States have showed an opposite pattern, with low historical catches that increased in the 1980s (largely due to the introduction of recreational monitoring). Catches in the Western Central EEZs have increased gradually over time.

```
# remove other oceans
head(tab_glob)
```

	other oceans	Mediterranean Sea EEZs	United States EEZ	Brazil EEZ
1950	19331387	26162262	1548245	1127242
1951	21573930	25410830	1721403	1060734
1952	22569230	30206350	1871779	1128992
1953	25904815	31275619	2024535	1111008
1954	32505789	31383840	2177865	1090492
1955	34571430	30056210	2309815	1190424

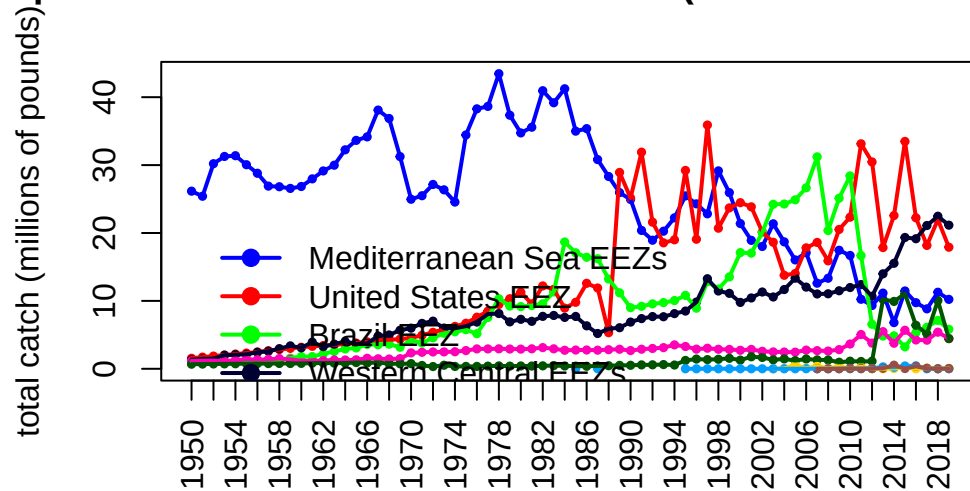
	Western Central EEZs	Northeast EEZs	Eastern Central EEZs	Southwest EEZs
1950	1345027	1173441	677552.5	NA
1951	1467925	1187386	693004.4	NA
1952	1510406	1193045	708466.7	NA
1953	2001465	1141504	723939.4	NA
1954	2069754	1269523	739422.5	NA
1955	2130341	1405543	754915.9	NA

	Southeast EEZs	Canadian EEZ
1950	NA	NA
1951	NA	NA
1952	NA	NA
1953	NA	NA
1954	NA	NA
1955	NA	NA

```
tab <- tab_glob[, -which(colnames(tab_glob) == "other oceans")]
#head(tab)
```

```
matplot(yrs, tab/10^6, las = 2, type = "b", pch = 19, cex = 0.5, lty = 1, col = glasbey(ncol(tab)),
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Dolphin catches from the Atlantic EEZs (commercial + recreational)")
matplot(yrs, tab/10^6, las = 2, type = "l", pch = 19, lty = 1, lwd = 2, add = T, col = glasbey(ncol(tab)))
axis(1, at = yrs[seq(1, 90, 2)], las = 2); axis(2); box()
legend(1952, 22, colnames(tab), col = glasbey(ncol(tab)), pch = 19, lty = 1, lwd = 2, bty = "n")
```

olphin catches from the Atlantic EEZs (commercial + recrea



4.7 Merge high seas with catch with exclusive economic zones

Now we will merge in the high seas catch from the steps above into the EEZ catches to get a better picture of total catch in FAO zones 31 and 21.

```
hs <- read.csv("data/SAU/high_seas_by_year.csv")
names(hs)[1] <- "year"

# check that columns match
head(tab)
```

	Mediterranean Sea EEZs	United States EEZ	Brazil EEZ	Western Central EEZs
1950	26162262	1548245	1127242	1345027
1951	25410830	1721403	1060734	1467925
1952	30206350	1871779	1128992	1510406
1953	31275619	2024535	1111008	2001465
1954	31383840	2177865	1090492	2069754
1955	30056210	2309815	1190424	2130341
	Northeast EEZs	Eastern Central EEZs	Southwest EEZs	Southeast EEZs

4 Analysis of international landings estimates from Sea Around Us database

1950	1173441	677552.5	NA	NA
1951	1187386	693004.4	NA	NA
1952	1193045	708466.7	NA	NA
1953	1141504	723939.4	NA	NA
1954	1269523	739422.5	NA	NA
1955	1405543	754915.9	NA	NA

Canadian EEZ

1950	NA
1951	NA
1952	NA
1953	NA
1954	NA
1955	NA

```
head(hs)
```

	year	Atlantic..Northwest	Atlantic..Western.Central
1	1950	0	0
2	1951	0	0
3	1952	0	0
4	1953	0	0
5	1954	0	0
6	1955	0	0

```
table(rownames(tab) == hs$year)
```

```
TRUE
70
```

```
tab[which(is.na(tab))] <- 0
tab_atl <- cbind(tab, hs[2:3])

# check the merge
head(tab_atl)
```

	Mediterranean Sea EEZs	United States EEZ	Brazil EEZ	Western Central EEZs
1950	26162262	1548245	1127242	1345027

4 Analysis of international landings estimates from Sea Around Us database

1951	25410830	1721403	1060734	1467925
1952	30206350	1871779	1128992	1510406
1953	31275619	2024535	1111008	2001465
1954	31383840	2177865	1090492	2069754
1955	30056210	2309815	1190424	2130341

	Northeast EEZs	Eastern Central EEZs	Southwest EEZs	Southeast EEZs
1950	1173441	677552.5	0	0
1951	1187386	693004.4	0	0
1952	1193045	708466.7	0	0
1953	1141504	723939.4	0	0
1954	1269523	739422.5	0	0
1955	1405543	754915.9	0	0

	Canadian EEZ	Atlantic..Northwest	Atlantic..Western	Central
1950	0	0		0
1951	0	0		0
1952	0	0		0
1953	0	0		0
1954	0	0		0
1955	0	0		0

```
# separate all WCA catches
dwc <- d[which(d$reg == "Western Central EEZs" | d$reg == "United States EEZ" | d$reg == "Canadian EEZ")]

#unique(dwc$area_name[which(dwc$reg == "United States EEZ")])
#unique(dwc$area_name[which(dwc$reg == "Canadian EEZ")])
#unique(dwc$area_name[which(dwc$reg == "Western Central EEZs")])

# relabel U.S. by area
dwc$reg[which(dwc$area_name == "USA (East Coast)")] <- "United States East Coast"
dwc$reg[which(dwc$area_name == "USA (Gulf of Mexico)")] <- "United States Gulf of Mexico"
dwc$reg[which(dwc$area_name == "Puerto Rico (USA)")] <- "United States Caribbean"
dwc$reg[which(dwc$area_name == "US Virgin Isl.")] <- "United States Caribbean"
#table(dwc$area_name, dwc$reg)

tab_wc <- tapply(dwc$lbs, list(dwc$year, dwc$reg), sum, na.rm = T)
head(tab_wc)
```

	Canadian EEZ	United States Caribbean	United States East Coast
1950	NA	142562.8	1239020
1951	NA	168365.6	1381482

4 Analysis of international landings estimates from Sea Around Us database

1952	NA	168463.5	1501267
1953	NA	162135.2	1623956
1954	NA	168659.3	1744386
1955	NA	175830.1	1846063

	United States	Gulf of Mexico	Western Central	EEZs
1950		309225.3		1202464
1951		339921.3		1299559
1952		370512.4		1341943
1953		400578.5		1839330
1954		433479.6		1901095
1955		463752.0		1954511

```
# reorder S to N
tab_wc <- tab_wc[, c(5, 2, 4, 3, 1)]
head(tab_wc)
```

	Western Central	EEZs	United States	Caribbean	United States	Gulf of Mexico
1950		1202464		142562.8		309225.3
1951		1299559		168365.6		339921.3
1952		1341943		168463.5		370512.4
1953		1839330		162135.2		400578.5
1954		1901095		168659.3		433479.6
1955		1954511		175830.1		463752.0

	United States	East Coast	Canadian	EEZ
1950		1239020		NA
1951		1381482		NA
1952		1501267		NA
1953		1623956		NA
1954		1744386		NA
1955		1846063		NA

```
# merge in the high seas catch - check years match
head(hs)
```

	year	Atlantic..Northwest	Atlantic..Western.Central
1	1950	0	0
2	1951	0	0
3	1952	0	0
4	1953	0	0

4 Analysis of international landings estimates from Sea Around Us database

```
5 1954          0          0
6 1955          0          0
```

```
table(rownames(tab_wc) == hs$year)
```

```
TRUE
 70
```

```
high_seas <- hs$Atlantic..Western.Central + hs$Atlantic..Northwest
high_seas[high_seas == 0] <- NA
tab_wc <- cbind(tab_wc, high_seas)
head(tab_wc)
```

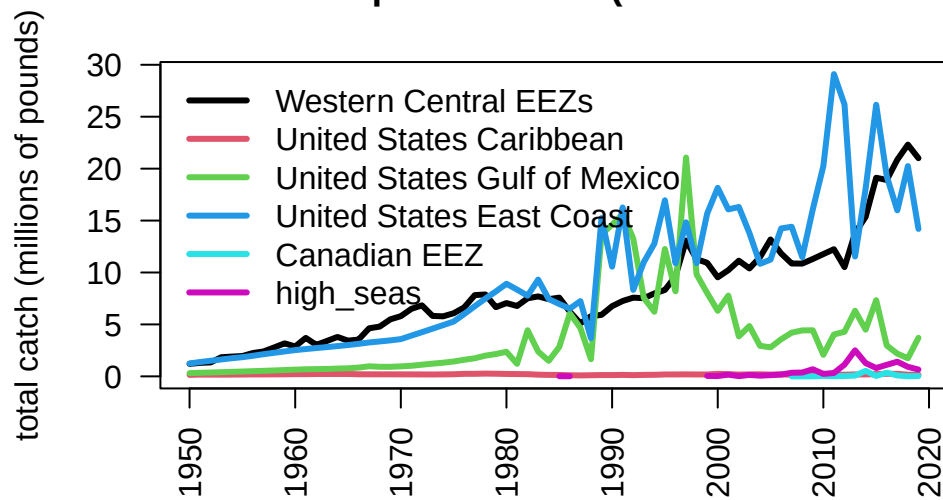
	Western Central EEZs United States	Caribbean United States	Gulf of Mexico
1950	1202464	142562.8	309225.3
1951	1299559	168365.6	339921.3
1952	1341943	168463.5	370512.4
1953	1839330	162135.2	400578.5
1954	1901095	168659.3	433479.6
1955	1954511	175830.1	463752.0

	United States East Coast Canadian EEZ	high_seas
1950	1239020	NA
1951	1381482	NA
1952	1501267	NA
1953	1623956	NA
1954	1744386	NA
1955	1846063	NA

4.8 Plot Western Atlantic catches

```
par(mar = c(5, 4, 4, 3), mgp = c(3, 1, 0))
matplot(yrs, tab_wc/10^6, las = 2, type = "l", lty = 1, lwd = 3, col = 1:6,
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Western Atlantic dolphin catches (commercial + recreational)")
legend("topleft", colnames(tab_wc), col = 1:6, lty = 1, lwd = 3, bty = "n")
```

Western Atlantic dolphin catches (commercial + recreational)



5 SST

6 Analysis of ocean temperature trends

Coming soon